Knowledge Tracing with Vision-Language Models

Two Approaches to Student Performance Prediction

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Problem Statement

Knowledge Tracing (KT) is the task of predicting whether a student will answer the next exercise correctly, given their history of past attempts. This work explores two experimental pipelines trained and evaluated on the **MIAAM dataset**, which contains 38,519 students with an average of 168 exercises each. Both pipelines leverage **SmolVLM-256M-Instruct** as the multimodal backbone and share a common data representation: exercises are described by text and a screenshot, while attempts are described by text (answer given, duration, timestamp, and correctness label).

The first experiment is intended as a **baseline**: feeding raw multimodal content directly to a VLM is the most straightforward approach to the problem, and it helps characterise the limitations of this strategy on the MIAAM dataset. The second experiment proposes an alternative design in which the VLM is used to encode each exercise and each attempt *independently* into a fixed-size vector; these representations are then consumed by a Transformer architecture, which can reason over much longer student histories.

Experiment 1 — Raw Multimodal Input

The first approach feeds *raw* multimodal content directly to the VLM at training time. For a student with history t, the model receives a concatenated sequence of exercise screenshots and text, together with text descriptions of past attempts.

Objective. Establish a baseline by fine-tuning a VLM either end-to-end or partially, directly on the raw multimodal input.

Two training regimes are supported:

- **Probe** — the VLM backbone is fully frozen; only a linear classification head appended to the pooled hidden state is trained. This is fast and parameter-efficient.
- **LoRA fine-tuning** — lightweight LoRA adapters are injected into the attention and MLP projections of the backbone and trained end-to-end alongside the head, at a lower learning rate than the probe.

Key limitations. The 256M-parameter VLM has a small context window, which restricts each input to only 4–5 past attempts regardless of the student's actual history length. Since students average 168 exercises, the vast majority of each student's learning trajectory is discarded at every training step, inherently limiting what the model can infer about long-term knowledge state.

A second limitation is **training speed**. Even in probe mode, where the backbone is frozen and no gradients flow through it, a full VLM forward pass must run on every batch to produce the hidden states consumed by the classification head. Processing raw screenshots through the vision encoder at every step is expensive, and this cost is paid repeatedly across all epochs. By contrast, Experiment 2 pays the VLM inference cost only once during preprocessing: embeddings are cached to disk and the training loop never touches the VLM again, making each training step orders of magnitude cheaper.

Experiment 2 — Pre-computed Embeddings

The second approach decouples the VLM from the training loop entirely. Each exercise and each attempt is encoded *once* by a frozen VLM into a 576-dimensional embedding vector. A lightweight Transformer then learns to reason over sequences of these cached embeddings.

Objectives. (1) Validate the advantage of pre-computing exercise and attempt embeddings from a computational perspective, expecting faster training that can run on smaller GPUs. (2) Validate the advantage in terms of prediction performances of feeding the model a longer sequence of student history.

Input sequence. For a student who has completed t exercises, the model receives an interleaved sequence of $2t + 1$ vectors:

$$[\, e_0,\, a_0,\, e_1,\, a_1,\, \ldots,\, e_{t-1},\, a_{t-1},\, e_t \,]$$

where $e_i = \mathrm{VLM}(\text{exercise}_i)$ and $a_i = \mathrm{VLM}(\text{attempt}_i)$. The final token e_t has no corresponding attempt yet; the prediction head reads the Transformer's hidden state at that position and produces a scalar logit for binary classification.

Architecture. The `KTTransformer` consists of (i) a linear projection from the 576-d input space to the model's internal dimension d_{model}, (ii) learnable positional embeddings, (iii) a stack of standard Transformer encoder layers (bidirectional attention, no causal mask), and (iv) a two-layer MLP head. Sequences are zero-padded to the batch maximum and a boolean attention mask prevents the model from attending to padding tokens.

Key advantage. Because embeddings are pre-computed and compact, the Transformer can process histories of up to **512 past exercises**, covering most students in full rather than truncating their history.

Results

No complete results was found for Experiment 1. A single training epoch on the available GPU takes 3–4 hours due to the cost of running the full VLM forward pass at every step, making a full training run impractical within the available time budget.

The embedded approach reaches **83% validation accuracy** after a short training session, with no hyperparameter tuning. This is a strong baseline given the simplicity of the encoder (256M parameters) and the absence of any hand-crafted features.

Conclusion

Experiment 1 lets the VLM jointly encode each exercise and reason over the full history end-to-end, so it can interpret an exercise in light of what just happened (e.g. right after a failure vs. a success), at the cost of higher capacity demands and the risk of entangling exercise content with spurious features of any one student's trajectory. Experiment 2 factorizes the problem — a frozen encoder produces consistent, context-free exercise embeddings that a dedicated downstream transformer contextualizes over the sequence — which is more parameter-efficient and avoids baking history-specific noise into the content representation, but loses the ability to fuse context at encoding time if the embedding discards the relevant detail.

Regarding Experiment 1, it would still be interesting to run a complete training on this setup, potentially using a more powerful model with a larger context window to better capture student history.

Experiment 2 appears very promising, especially when one of the main desiderata of the project is to operate with small VLMs. It offers clear advantages in training speed and in the ability to feed the model a much longer student history. The natural next step here is to try a slightly larger backbone to produce richer embeddings, while retaining the decoupled, computationally efficient design.